Dataset Description

Specification of Synthetic AUTOSAR High-Level Design Benchmark

1. Dataset Overview & Benchmark Rationale

To validate the analysis assistant without compromising confidential automotive intellectual property, a comprehensive synthetic AUTOSAR Classic Release 4.4.0 High-Level Design document was authored: 'Sample_AUTOSAR_HLD.pdf'.

The document specifies a dual-motor electric Powertrain and Chassis Domain Controller (PCDC) coordinating electric traction, anti-lock friction braking, and high-voltage battery management.

2. Specification Structure & Page Breakdown

The synthetic HLD spans 12 structured technical pages:

• Page 1: Document Classification, Classification Notice, and Powertrain Scope

• Page 2: AUTOSAR Software Layering, Virtual Functional Bus, and SWC Inventory

• Page 3: VehicleSpeedSWC (SensorSWC, ASIL-B, Kalman filtering, 10ms runnable)

• Page 4: BrakeControlSWC (ApplicationSWC, ASIL-D, ABS & brake blending arbitration)

• Page 5: MotorControlSWC (ApplicationSWC, ASIL-D, inverter torque & regen control)

• Page 6: BatteryManagementSWC (ApplicationSWC, ASIL-C, 800V pack monitor, thermal limits)

• Page 7: DiagnosticManagerSWC (ServiceSWC, ASIL-B, UDS ISO 14229 fault logging)

• Page 8: Sender-Receiver Port Interfaces Specification (If_VehicleSpeed, If_BrakeTorqueCommand, etc.)

• Page 9: Client-Server Operations Specification (If_DiagnosticService, If_PowerDerateRequest)

• Page 10: Signals and Global Data Dictionary (10 structured signal elements with units and types)

• Page 11: Architectural Dependencies and Regenerative Braking Functional Flow

• Page 12: RTE Task Mapping (5ms, 10ms, 50ms, 100ms), MPU Partitioning, and ISO 26262 Requirements

3. Component & Signal Ground Truth Summary

Component Name	Category	ASIL	Key Interface	Periodicity
VehicleSpeedSWC	SensorSWC	ASIL-B	If_VehicleSpeed (S-R)	10ms
BrakeControlSWC	ApplicationSWC	ASIL-D	If_BrakeTorqueCommand (S-R)	10ms
MotorControlSWC	ApplicationSWC	ASIL-D	If_MotorTorque (S-R)	5ms
BatteryManagementSWC	ApplicationSWC	ASIL-C	If_BatteryState (S-R)	50ms
DiagnosticManagerSWC	ServiceSWC	ASIL-B	If_DiagnosticService (C-S)	100ms